Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

INSULATION TESTER · HAND-DRIVEN · METAL BODY

CIE/444

Hand-driven insulation tester in anti-magnetic aluminium

Direct-reading insulation resistance testing

A compact, portable instrument for direct reading of insulation resistance on electrical operating machines, rotating plant, stationary apparatus, insulation control and switching equipment, distribution networks, communication cables, insulators, transformers, industrial installations, cables, wiring and other devices capable of carrying the rated voltage. It is a versatile instrument for locating intermittent shorts, defective electrical connections, insulation failure and conductor failure, and for preventing costly repairs and breakdowns.



CIE/444 · ILLUSTRATIVE IMAGE

27	6	3 s
STANDARD RANGES	RATED VOLTAGES	RESPONSE TIME

STANDARD RATED TEST VOLTAGES

100 V · 250 V · 500 V · 1,000 V · 2,500 V · 5,000 V

01 HIGHLIGHTS

- ✓ Anti-magnetic aluminium body
- ✓ Silent generator
- ✓ Collapsible driven handle, folds back flush into the casing
- ✓ Wide anti-parallax mirror scale
- ✓ Long life and strength against vibration and impact
- ✓ Guard terminal and externally accessible infinity adjuster on ranges of 1000 MΩ and above
- ✓ Protected metallic flap over the scale window
- ✓ Centrifugal clutch (governor) for constant output voltage
- ✓ Chuck nut locking arrangement prevents anti-clockwise rotation
- ✓ Highly sensitive cross-coil indicator
- ✓ Accurate and quick response (3 seconds)
- ✓ Conforms to IS 2992:1987

02 TYPICAL APPLICATIONS

- Motors, generators, and other rotating machinery
- Insulation control and switching equipment checks
- Distribution network and communication cable insulation testing
- Transformer and insulator testing; industrial installation checks
- Locating intermittent shorts, defective connections, and developing insulation failures

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03 HOW IT WORKS

PRINCIPLE

A hand-driven clutch-type A.C. generator and a highly sensitive cross-coil indicator. The generated A.C. voltage is converted to D.C. by silicon diodes.

CONSTRUCTION

Housed in a sturdy, elegant case of anti-magnetic cast aluminium with a metallic carrying handle. The instrument is dust-free: rubber shoes hold the surface and give it stability, and a protected metallic flap covers the glass window of the scale.

04 SALIENT FEATURES

Silent generator

The A.C. generator runs extremely smoothly and silently thanks to the dynamic balance of its rotating parts, and is free from mechanical noise.

Centrifugal governor

A high-grade centrifugal governor holds the voltage constant for accurate measurement. Variation in crank speed does not affect the reading, and a cut-off clutch disengages the drive beyond the rated speed.

Collapsible handle

The generator handle folds back flush into the casing when the tester is not in use.

Quick response

3 seconds, when a resistance equal to the central scale mark is suddenly inserted between the measurement terminals, as per IS 2992:1987.

05 TECHNICAL SPECIFICATIONS

Body	**Anti-magnetic aluminium**
Generator Type	**Hand-driven, silent generator with centrifugal governor (clutch)**
Scale Length	**90 mm (approx.)**
Accuracy	**Conforms to IS: 2992-1987**
Rotation Speed	**160 r.p.m. or more**
High Voltage Test	**Withstands minimum 2000 V or (rated voltage + 1000) V A.C. (rms) between electrical circuit and handle, for 1 minute**
Insulation Resistance Test	**Better than 50 MΩ at 500 V between electrical circuit and containing case**
Guard Terminal & Infinity Adjuster	**Provided on instruments with insulation range 1000 MΩ and above**
Response Time	**3 seconds**
Weight	**2.5 kg (approx.)**
Dimensions (L×W×H)	**110 × 128 × 195 mm (approx.)**

Ranges are listed on the following pages. Specifications are approximate where marked.

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06 STANDARD RANGES BY RATED VOLTAGE

RATED VOLTAGE	INSULATION RESISTANCE RANGE (MΩ)
100 V	0–20 · 0–50 · 0–100
250 V	0–50 · 0–100 · 0–250
500 V	0–100 · 0–200 · 0–500 · 0–1,000 · 0–2,000
1,000 V	0–100 · 0–200 · 0–500 · 0–1,000 · 0–2,000
2,500 V	0–1,000 · 0–2,000 · 0–2,500 · 0–5,000 · 0–10,000 · 0–20,000 · 50–50,000
5,000 V	0–5,000 · 0–10,000 · 0–20,000 · 50–50,000

The 100,000 MΩ range at 5,000 V is not offered on the metal-body CIE/444 (available on the wooden-body CIE/666).

07 COMBINING VOLTAGES AND RANGES

Dual voltage & dual insulation range

One instrument, two rated voltages, each with its own insulation range, selected by a switch. The selected range must lie within the rated resistance listed for that voltage (voltage to measurable insulation range ratio up to 1:2 only; for any other ratio, enquire). Not available for the 10,000 V instrument.

Single voltage, dual insulation range

One rated voltage with two insulation ranges. Both ranges must lie within the rated resistance listed for that voltage. Not available for the 10,000 V instrument.

Continuity range (500 V and 1,000 V only)

Built in as an optional extra on single-voltage, single-range instruments, or on dual-voltage, dual-range instruments.

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08 MULTIRANGE SERIES: CIE/444/MR

SERIES	RATED VOLTAGE / INSULATION RANGE (MΩ)	CONTINUITY
MR-1	1,000 V / 0–2,000 500 V / 0–1,000 250 V / 0–500	0–20 kΩ
MR-2	1,000 V / 0–200 500 V / 0–100 250 V / 0–50	0–1 kΩ
MR-3	500 V / 0–1,000 250 V / 0–500 100 V / 0–200	0–5 kΩ
MR-4	500 V / 0–100 250 V / 0–50 100 V / 0–20	0–1 kΩ

Multirange instruments carry a range selector switch, so one instrument covers every voltage and range listed in its series. Anti-parallax mirror scale is not provided on triple-range and above instruments. Any other configuration can be built to order.

09 OPTIONAL ACCESSORIES (AT EXTRA COST)

- Carrying case (metal body)
- Testing leads, 1 m or 2 m (up to 1,000 V)
- H.V. testing leads, 2 m (general / special)

10 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example **CIE/444, 500 V / 0–2,000 MΩ**. Any range not listed can be manufactured against specific requirements.

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